AM and FM Simulation Code

Python code for Google Colab / Jupyter Notebook

Purpose: Generate and plot the message signal, carrier signal, AM waveform, and FM waveform.

```
import numpy as np
import matplotlib.pyplot as plt

# 1. Signal parameters
Am = 1.0       # Message amplitude
Ac = 1.0       # Carrier amplitude
fm = 5         # Message frequency (Hz)
fc = 50        # Carrier frequency (Hz)
fs = 5000      # Sampling frequency (Hz)
duration = 1   # Duration in seconds

t = np.arange(0, duration, 1/fs)

# 2. Message signal
message = Am * np.cos(2 * np.pi * fm * t)

# 3. Carrier signal
carrier = Ac * np.cos(2 * np.pi * fc * t)

# 4. AM modulation
mu = 0.8  # Modulation index
am_signal = Ac * (1 + mu * message) * np.cos(2 * np.pi * fc * t)

# 5. FM modulation
beta = 5  # FM modulation index
fm_signal = Ac * np.cos(
    2 * np.pi * fc * t
    + beta * np.sin(2 * np.pi * fm * t)
)

# 6. Plot all signals
plt.figure(figsize=(12, 10))

plt.subplot(4, 1, 1)
plt.plot(t, message)
plt.title("Message Signal")
plt.ylabel("Amplitude")
plt.xlim(0, 0.5)
plt.grid(True)

plt.subplot(4, 1, 2)
plt.plot(t, carrier)
plt.title("Carrier Signal")
plt.ylabel("Amplitude")
plt.xlim(0, 0.1)
plt.grid(True)

plt.subplot(4, 1, 3)
plt.plot(t, am_signal)
plt.title("Amplitude Modulation (AM)")
plt.ylabel("Amplitude")
plt.xlim(0, 0.5)
plt.grid(True)

plt.subplot(4, 1, 4)
plt.plot(t, fm_signal)
plt.title("Frequency Modulation (FM)")
plt.xlabel("Time (seconds)")
plt.ylabel("Amplitude")
plt.xlim(0, 0.5)
plt.grid(True)

plt.tight_layout()
plt.show()
```

How to run: Open Google Colab, create a new notebook, paste the code into one code cell, and press Run. The notebook will display four graphs.

Note: This is an AM/FM simulation. It does not by itself implement an AI feature; only claim AI functionality if it is separately included in your project.